Documenting AI in Research

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Open Research in the Digital Humanities

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AI as a Tool or a Methodological Component

AI can play two roles in research:

As a tool

It supports an existing research activity without fundamentally changing the method — for example, generating a first version of code, organising data, or assisting with data processing.

As a methodological component

It is integrated into the research design and affects how sources are processed and how data are created and analysed — for example, producing transcriptions, classifications, annotations, or extractions that shape the research material.

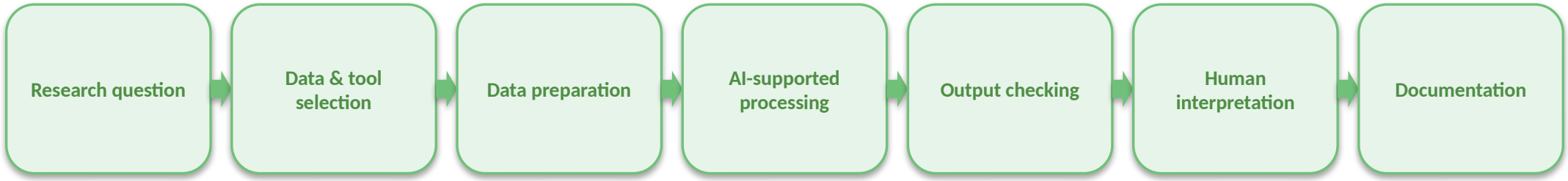
The same system can be a tool in one part of a project and a methodological component in another. Documentation should show at what stage and how AI contributed to producing knowledge.

References for further reading:

Oberbichler, Sarah, and Cindarella Petz. “Working Paper: Implementing Generative AI in the Historical Studies.” Zenodo, 2025. https://doi.org/10.5281/zenodo.14924737.

The Documented AI Pipeline

Documentation should describe the workflow and make the use of AI transparent — recording the research task, the AI system and configuration, the input and output, the validation, the researcher's decisions and revisions, and any limitations.



This is not a linear process. Humanities research is often iterative: researchers may return to the sources, revise categories, change the prompt, or reject some AI-based procedures on ethical grounds.

References for further reading:

"Using AI for Humanities Research Software." Institute for Research Software. https://www.software.ac.uk/guide/using-ai-humanities-research-software.

Disclosure Is Not Documentation

Disclosure

states that AI was used — for example, “AI was used for generating annotations.”

Documentation

explains how that use affected the research. It records:

- the purpose of the AI use
- the role of AI
- the system, model, version and date
- the input and configuration
- the output
- the evaluation
- the researcher’s decisions and revisions
- the limitations

The level of detail should be proportionate to the importance of the AI procedure: a brief note may suffice for an unsuccessful exploratory test, while actions that structure the dataset need a fuller record.

Ethics, Limitations, and Responsibility

- Documenting AI also means documenting its limitations.
- **Relevant risks:** inaccurate or invented outputs (“hallucinations”), unequal performance across languages and historical periods, bias in categories, loss of context, mishandling of ambiguity, privacy and copyright problems, exposure of restricted materials, and changes to models over time.
- Shape AI-related documentation to include failed experiments, inconsistent outputs, manually corrected records, and other human-assisted work combined with AI-supported actions.
- Such documentation makes research procedures traceable and transparently demonstrates human responsibility.

References for further reading:

Humphries, James, Michael Townsen Hicks, and Joe Slater. “LLMs Bullshit by Design: A Reply to Licon.” Philosophy & Technology 39, no. 2 (2026): 98. https://doi.org/10.1007/s13347-025-01016-x.

Vibe Coding and the Course Project

- **Vibe coding:** using generative AI systems to create or modify code across different stages of research.
- Learners without coding experience are encouraged to use AI for bounded tasks — obtaining, transforming and analysing data — which lowers the barrier to a more computational approach. But the code should not be trusted automatically.
- Treat AI-generated code as never ready to use: generate it, inspect it and run it in small steps; if something is unclear or unexpected, stop, ask for an explanation, and revise before continuing.
- Document the prompts, explanations, code, checks and corrections. You need not preserve every failed attempt or minor edit — document enough for the development history to explain how the final code was produced and checked, and save the final version.

References for further reading:

Cwik, Jan Christopher. “Disclosure Is Not Documentation: An Open Science Framework for Documenting Generative AI Use in Scholarly Research and Publication Workflows.” Research Integrity and Peer Review 11 (2026): 47. https://doi.org/10.1186/s41073-026-00245-8.

Feng, Yebo, and Yang Liu. “A Visionary Look at Vibe Researching.” Version 2. Preprint, arXiv, 2026. https://doi.org/10.48550/arXiv.2604.00945. This is an emerging conceptual proposal, not an established research method.

Humphries, James, Michael Townsen Hicks, and Joe Slater. “LLMs Bullshit by Design: A Reply to Licon.” Philosophy & Technology 39, no. 2 (2026): 98. https://doi.org/10.1007/s13347-025-01016-x.

Ma, Rongqian, Xuhan Zhang, and Adrian Wisnicki. “Disclosing Generative AI Use in Digital Humanities Research.” Proceedings of the Association for Information Science and Technology 62, no. 1 (2025): 1572–74. https://doi.org/10.1002/pra2.1471.

Oberbichler, Sarah, and Cindarella Petz. “Working Paper: Implementing Generative AI in the Historical Studies.” Zenodo, 2025. https://doi.org/10.5281/zenodo.14924737.

Wills, Simon, Sanson T. S. Poon, Arianna Salili-James, and Ben Scott. “The Use of Generative AI for Coding in Academia.” Methods in Ecology and Evolution 15, no. 12 (2024): 2189–91. https://doi.org/10.1111/2041-210X.14454.